

Executive Summary

Product X (48-hour post-EVT infusion) addresses a clear unmet need in LVO stroke by potentially improving functional recovery and survival beyond EVT alone ¹ ² . To qualify for NTAP and gain payer support, Phase 3 must show **statistically significant, clinically meaningful gains** in outcomes (e.g. $\geq 15\text{--}20\%$ absolute increase in mRS 0–2 and proportional mortality reduction, with $p < 0.05$) vs EVT alone. Trial design should target the post-EVT LVO population (likely including older/Medicare-age patients), with endpoints like 90-day mRS, Barthel Index, mortality, and adverse events. Safety must be acceptable (no excess serious AEs, bleeding) and infusion tolerability established. Evidence should align with CMS’s “substantial clinical improvement” (SCI) criteria – i.e. reduce mortality/complications and improve ADLs/quality of life ¹ ³ .

The coding landscape is complex. Inpatient claims use **ICD-10-PCS** (not HCPCS) to identify the infusion. Likely candidates are existing codes for “Introduction of other therapeutic substance into (peri- or central) vein” (e.g. **3E033GC**, **3E043GC**). However, to ensure visibility, we recommend requesting one or more **new ICD-10-PCS codes in Section X** (New Technology) specifically for this prolonged infusion. HCPCS/J-codes (for drugs) are not needed for NTAP but may be pursued for outpatient billing. Standard inpatient revenue codes (e.g. pharmacy 025X, IV therapy 026X) will capture the charge.

Product X’s value comes from **better outcomes and cost offsets**: Phase 2 data project ~3–5 fewer hospital days, a 15–30% relative boost in discharge-to-home rates, and substantial mortality and ADL improvements. For example, a 4-day LOS reduction (~\$8K at ~\$2K/day) and higher home discharge (~20% more) could cut downstream rehabilitation/SNF costs by thousands. These gains translate to QALYs: more survivors in mRS 0–2 yields higher utilities. Detailed cost-benefit modeling (not shown) would quantify these offsets, but qualitatively the added value is strong.

For coding strategy, we will submit an **ICD-10-PCS new code request** (targeting 2025-Q4 for FY2027 NTAP). We will also plan for a **new HCPCS (J-code)** application after FDA approval if clinically justified (useful for Part B coverage or commercial claims) and consider CPT/HCPCS for infusion administration if needed (though inpatient use relies on the ICD-10 code and revenue codes). The ICD-10 code submission should follow CMS guidance on timeline (submit by December 2025 for FY2027 NTAP) ⁴ .

In modeling NTAP payment, at a baseline DRG pay ($\approx \$30\text{K--}35\text{K}$) the infusion’s cost is incremental. NTAP would cover 65% of that cost ⁵ . For instance, at a **\$30K drug price**, NTAP ~19.5K (65%), leaving ~10.5K as hospital’s incremental cost. We summarize scenarios below:

Price (\$)	NTAP (65%)	Hospital Pays (35%)
20,000	13,000	7,000
30,000	19,500	10,500
40,000	26,000	14,000
50,000	32,500	17,500

(Hosp. pays = price – NTAP)

Based on CMS precedent, Product X's NTAP chance is **moderate-high**. It is unquestionably “new” and clearly costly; with strong Phase 3 outcomes it likely meets SCI. There is no directly competing tech (differentiation risk is low), and having a unique procedural profile (48h ICU infusion) improves visibility. Risks are chiefly evidentiary: Phase 3 must confirm Phase 2 trends. Without robust Phase 3 data, NTAP (and payer) approval would be less likely.

Action Plan: Emphasize robust functional/mortality endpoints (e.g. mRS and Barthel), power the trial adequately, and include Medicare-age patients. Collect resource use (LOS, disposition) prospectively. Immediately prepare an ICD-10-PCS code application (target Section X code(s)), and schedule NTAP pre-submission (NTAP due Oct 2025, town hall Dec 2025 ⁴). Engage payers (PIE messaging) early to cement the \$30K+ add-on narrative (leveraging Aetna's willingness). Seek HCPCS code post-approval for Part B coverage. The timeline below coordinates these parallel tracks.

1) Phase 3 Evidence Requirements (SCI and Payer Expectations)

- **Primary Endpoint:** Functional outcome (e.g. mRS 0–2 at 90 days). Phase 2 projected a **13–16% absolute improvement** over placebo. Phase 3 should target a **≥15% absolute increase** with *statistical significance* (typically $p < 0.05$) to meet CMS SCI ¹ ². Ideally, the primary endpoint is a binary functional outcome; alternatives could include shift analyses or $BI \geq 75$.
- **Secondary Endpoints:** 90-day **mortality**, **Barthel Index improvement**, NIHSS change, discharge disposition (home vs facility), and quality-of-life metrics (e.g. EQ-5D for QALYs). The CFR explicitly values **reduced mortality or serious complications** ¹ and **improved ADLs/quality of life** ³. We should aim to show a significant mortality reduction (~15–30% relative as in P2) and significant gains in ADL scores.
- **Safety:** Demonstrate *no increase* in serious AEs (e.g. ICH, cardiac events). Transient, mild AEs (e.g. headache) are acceptable. Safety data must support a favorable risk–benefit.
- **Population:** Patients with acute LVO stroke undergoing EVT (e.g. within 6–12h of onset). Include broad representation (≥ 65 , comorbidities) to address Medicare demographics, as CMS wants data generalizable to its population ². Consider pre-specifying subgroups (e.g. extended window) if relevant.
- **Design:** Randomized, double-blind, placebo-controlled (control = standard post-EVT care). Adequate sample size to detect the above effect (likely several hundred patients). Include cost-utility endpoints collection (resource use, QoL utilities). Ensure robust data capture (central core lab imaging, functional assessments by certified raters).
- **Statistical Thresholds:** Two-sided alpha 0.05 for primary; power ≥ 80 –90%. A prespecified hierarchical testing of secondary outcomes can strengthen evidence. Noninferiority margins for safety endpoints (e.g. mortality) should ensure no worse outcomes.
- **Payer Considerations:** Payers will scrutinize if outcomes are clinically meaningful and cost-offsetting. Strong evidence on functional independence and reduced downstream care (e.g. less rehab/SNF due to home discharge) will be persuasive. They will expect real-world applicability (e.g. minimal exclusion criteria).

2) Coding & Reimbursement Landscape (ICD-10, HCPCS, CPT, Revenue)

- **ICD-10-PCS (Inpatient Code):** Inpatient claims use ICD-10-PCS. The generic codes for a prolonged IV infusion are:
 - 3E033GC – Introduction of *Other Therapeutic Substance* into Peripheral Vein, Percutaneous.
 - 3E043GC – Introduction of *Other Therapeutic Substance* into Central Vein, Percutaneous.
 These could be used, but they group **all** 48h infusions of any “other substance.” To ensure the infusion is **uniquely identifiable**, a new ICD-10-PCS code (Section X) is recommended. For example, Section X code(s) could be requested to reflect “Product X infusion for 48h”. (See **Table: ICD-10-PCS Options** below.) We should apply for an ICD-10-PCS code by Dec 2025 (in time for Oct 2026 NTAP) ⁴.
- **ICD-10-PCS Code Request:** Strategy is to propose a new code under the Medical/Surgical section “Introduction” to (peri-/central) vein, qualifying by a new identifier for Product X. CMS usually classifies new drugs under “New Technology” (Section X, Body System W, Operation X, etc.) with generic “New Technology Group” qualifiers (8,9). We will draft a request emphasizing the unique 48h protocol and track usage for NTAP. The existing codes (3E033GC/3E043GC) may serve short-term if a new code isn’t yet issued, but ultimately a new code is best. CMS guidance notes that codes can be submitted separately and must be timely ⁴.
- **HCPCS/J-Codes:** HCPCS (J-codes) primarily govern outpatient drug billing. For inpatient NTAP, they aren’t required. However, applying for a new J-code (before or after approval) could aid commercial or outpatient coverage. If pursued, it would be after FDA approval (AMA/Palmetto process) and would allow each IV bag to be charged on claims. Alternatively, hospitals might temporarily bill under existing J2990 (unclassified drug) for data collection, but a distinct J-code would be preferable for tracking.
- **CPT/E&M Codes:** CPT codes apply to physician services and outpatient infusions. Inpatient infusion is not reported with CPT. For completeness: if used in an outpatient setting (unlikely in acute stroke), the infusion could use 96365/96366 (pharmacologic infusion codes). For evaluation, hospital-employed physicians bill usual E&M codes (99221–99233). We do not anticipate needing new CPT codes specific to Product X beyond standard infusion administration codes for any outpatient use.
- **Revenue Codes:** Hospital chargemasters should allocate Product X costs to pharmacy and IV-therapy revenue centers. Likely codes: **025X** (Pharmacy, e.g. 0252 Non-Generic drugs) and **026X** (IV Therapy; 0264 supplies or 0260 general) for the IV supplies/pump. The main charge will be captured under pharmacy (025) and/or IV therapy (026). This ensures the drug cost is visible on the UB-04, which NTAP uses in its cost-threshold analysis ⁴.
- **Charge Capture:** The hospital must bill each 48h infusion explicitly (one or multiple units) on the claim. For NTAP, CMS uses the claim’s charges to compute the technology’s cost. We should ensure clear usage of the new (or existing) code on claims and track unit counts (e.g. X bags per case).

Table: ICD-10-PCS Code Options (draft)

Code (tentative)	Descriptor	Comments
3E033GC	Intro Other Subst into Peripheral Vein, Perc. (Existing)	Covers any 48h IV; non-specific.
3E043GC	Intro Other Subst into Central Vein, Perc. (Existing)	For central-line infusions.
XW0X3GA	<i>New Technology</i> code: Intro of ProdX into Periph. Vein, Perc, New Technology Group 8	Proposed to identify Product X infusion specifically.
XW0X4GA	<i>New Technology</i> code: Intro of ProdX into Central Vein, Perc, New Technology Group 8	Optional if infusion often via central line.

Note: New codes (Section X, Body W) would use generic “Other Substance” with a “New Technology” qualifier. We would draft code titles reflecting 48h infusion. Submission by Dec 2025 is crucial ⁴.

3) Treatment’s Value & HEOR Impact

- **Improved Clinical Outcomes:** Based on Phase 2/TPP data, Product X can yield a **15–30% relative mortality reduction** and ~13–16% absolute increase in favorable mRS [TPP]. Such gains significantly improve patient survival and independence. More patients returning home (15–30% higher rate) reflects better quality of life and reduces demand for costly post-acute care (e.g. rehab, SNFs). These translate into higher QALYs (fewer years with disability), a key HEOR metric. (For example, each additional patient in mRS 0–2 may gain ~0.7 utility vs ~0.2 for mRS 3–5.)
- **LOS/ICU Reduction:** Phase 2 suggests ~3–5 fewer hospital days (with ~2–3 fewer ICU days) [TPP]. At an estimated **\$2,000–3,000 per hospital day** (higher for ICU), a 4-day saving is ~\$8K–12K cost avoided per case. Cumulatively, this offsets part of the drug cost. Faster discharge also frees beds. CMS explicitly recognizes “more rapid resolution of disease (reduced length of stay)” as an SCI factor ⁶.
- **Downstream Offsets:** Higher home discharge implies less inpatient rehab or SNF use, which can cost \$20–\$30K per episode. For example, if 20% more patients go directly home, many expensive rehab stays are avoided. Over 90 days, readmissions may also fall with better recovery. These reduced downstream costs strengthen a budget-impact argument.
- **HEOR Metrics:** A full cost-effectiveness analysis would compare the additional drug cost to QALYs gained. Roughly, preventing one severe disability could yield several QALYs at standard utility weights, often justifying tens of thousands of dollars per QALY. Even without formal modeling, the narrative is that improving mRS by ~15 points and cutting mortality by ~20% likely yields a favorable cost/QALY given stroke’s high baseline morbidity. Payers will expect at least a compelling projection of cost offsets and QALY gains (ideally via a BIA/CER model in PIE/HTA materials).

4) Code Application & Submission Strategy

- **ICD-10-PCS Submission:** Begin drafting in mid-2025. Submit to CMS by the December 2025 deadline for FY2027 consideration ⁴. We should prepare for an ICD-10 C&M presentation (likely March 2026) advocating new code(s) as outlined above. The request will note that existing codes are too generic and that Product X’s unique 48h protocol deserves distinct tracking (crucial for NTAP cost analyses).
- **HCPCS/J-Code Strategy:** Plan to file for a new J-code after FDA approval (target Jan 2027 if on-time). The J-code application would rely on the FDA label (e.g. “one 500ml bag per 8 hrs”). This would allow the hospital pharmacy to charge each bag under a defined J-code (useful for outliers/Medicare Part B or commercial billing). While not needed for NTAP, an approved J-code simplifies accounting and can be listed on UB-04 (Medicare doesn’t pay by ASP in inpatient, but commercial payers might).
- **CPT Codes:** No new CPT codes are anticipated solely for inpatient infusion. In outpatient infusion clinics, existing HCPCS 96365/96366 cover infusion. Physician billing will use standard E&M codes; no special add-on.
- **Timing Sequence:**
 - **NTAP Prep (Aug–Oct 2025):** Finalize evidence summary and cost threshold analysis; prepare MEARIS application (due Oct 2025) ⁷.
 - **ICD-10 Application (Nov–Dec 2025):** Submit code request by Dec 5, 2025 (for FY2027 NTAP) ⁴.
 - **PIE Engagement (2025–26):** Use interim data to educate payers (e.g. Aetna) via pre-approval meetings (per FDA PIE guidance ⁸) and secure coverage signals (like the ~\$30K add-on reported).

- **NTAP Town Hall (Dec 2025):** Possibly present SCI evidence to CMS (applications can present to CMS/public ⁹).
- **Proposed Rule (Apr 2026):** CMS posts summaries; prepare comments.
- **HCPCS Application (post-FDA, e.g. Q3 2026):** File with AMA/Palmetto for a new J-code to be effective Jan 2027.

5) NTAP Payment Modeling

CMS pays **the lesser of 65% of cost or 65% of excess DRG** ⁵. For simplicity, assume NTAP = 65% of Product X cost (since cost is well above DRG threshold). With a base MS-DRG (e.g. DRG 023/024 for stroke w/ EVT) covering roughly \$30–35K, the incremental Product X cost is added on. The table above illustrates per-case NTAP vs hospital share at different prices. For example, at a \$30K price, NTAP ≈ \$19.5K and the hospital covers the remaining ~\$10.5K. This supplemental payment can eliminate most of the negative margin. Sensitivity: if the DRG payment is actually lower (say \$25K), the incremental gap is larger and NTAP covers even more shortfall; if the infusion price is higher (>\$50K), the hospital burden grows (unless DRG is reweighted). After NTAP expiration, full cost falls on DRG, underscoring the importance of aligning on a higher DRG weight.

6) NTAP Approval Likelihood & Risks

We assess **moderate-high probability** of NTAP approval if Phase 3 succeeds. Rationale: (1) *Newness*: Product X is first-in-class for post-EVT stroke, satisfying “new” (within 2–3 years of launch) criterion ². (2) *Cost*: The 48h ICU infusion is clearly costly; average charges will exceed the MS-DRG payment threshold (costs exceed Medicare DRG for stroke) ⁵. (3) *SCI*: If Phase 3 replicates Phase 2 signals (notably lower mortality and better functional outcomes), it will meet the SCI bar via multiple CFR categories (mortality reduction, improved ADLs, shorter recovery) ^{1 3}. There is no existing comparator technology; the mechanism (mitochondrial protection) is unique for stroke, minimizing “substantially similar” risk.

Failure modes: Without strong Phase 3 significance or if FDA approval is delayed beyond May 1 (missing NTAP window ¹⁰), chances drop. Also, if CMS deems the improvements incremental (e.g. <5–10% mRS gain) or attributable mainly to patient selection, NTAP could be denied. A major risk is coding: if we have to use generic codes, CMS might say “no specific intervention; already in DRG.” Hence the emphasis on unique coding.

7) Recommendations & Action Plan (Prioritized)

- **Enhance Endpoints:** Ensure the Phase 3 protocol includes both NIHSS and patient-centered endpoints (mRS and BI). Consider adding cognitive or quality-of-life scales to strengthen HEOR arguments (e.g. EQ-5D for utility). Include 90-day and possibly 1-year follow-up if feasible.
- **Inclusion Criteria:** Minimize exclusions. Include broader age range and typical comorbidities to mirror real-world stroke patients (to satisfy Medicare applicability ²).
- **Collect Resource Data:** Prospectively track LOS (ICU vs ward), rehab days, readmissions. Use these to populate budget-impact and cost-effectiveness models.
- **Coding Steps:** File ICD-10-PCS code request immediately. Simultaneously, prepare request for any related CPT/HCPCS (especially J-code) applications so they can follow promptly after FDA ok.
- **PIE Messaging:** Develop payer-facing materials emphasizing the Phase 3 design, anticipated effect sizes, and budget impact. Leverage the Aetna \$30K signal as proof of payer willingness. Engage

major payers (e.g. Medicare Advantage plans, large self-insured) via PIE meetings to build demand. Ensure messaging aligns with NTAP evidence (don't overpromise).

- **NTAP Submission:** Prepare NTAP application mid-2025 with clinical summary, cost analysis, and coding plan. Respond to CMS feedback via the NTAP Town Hall and public comments if invited.
- **Timeline:** See below for a coordinated Gantt chart of activities. Key milestones: Phase 3 completion by mid-2026; NTAP Town Hall Dec 2025; Final NTAP decision Aug 2026.

Timeline: The mermaid Gantt chart below outlines parallel Phase 3 trial execution, coding/NTAP prep, PIE/ payer outreach, and HCPCS efforts spanning 2024–2026.

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gantt
    dateFormat YYYY-MM
    axisFormat %b '%y
    section Phase 3 Trial
    Design & Start Enrollment      :a1, 2025-01, 6m
    Continue Enrollment & Followup :a2, 2025-07, 18m
    Data Analysis & Toplines      :a3, 2027-01, 4m
    section NTAP & Coding
    NTAP Dossier Prep (MEARIS)    :b1, 2025-08, 3m
    ICD-10 Code Request (Design)  :b2, 2025-09, 3m
    Submit NTAP App (FY2027)      :b3, 2025-10, 0m
    Submit ICD-10 Code Request    :b4, 2025-12, 0m
    NTAP Town Hall Prep           :b5, 2025-11, 1m
    NTAP Town Hall Presentation    :b6, 2025-12, 0m
    ICD-10 C&M Committee Meeting  :b7, 2026-03, 0m
    section Payer Engagement (PIE)
    Pre-Approval Payer Meetings   :c1, 2025-07, 6m
    Incorporate Feedback to Value Proposition :c2, 2026-01, 6m
    File HCPCS (J-code) Application :c3, 2026-06, 2m
    section Regulatory
    FDA Submission for Approval    :d1, 2025-11, 7m
    Expected FDA Approval         :d2, 2026-06, 0m
    NTAP FDA Cut-off (May 1, 2026) :milestone, 2026-05, 0m
    IPPS Final Rule (NTAP final)  :d3, 2026-08, 0m
```

1 3 6 42 CFR § 412.87 - Additional payment for new medical services and technologies: General provisions. | Electronic Code of Federal Regulations (e-CFR) | US Law | LII / Legal Information Institute
<https://www.law.cornell.edu/cfr/text/42/412.87>

2 7 How a New Technology Add-On Payment (NTAP) Works | Avalere Health Advisory
<https://advisory.avalerehealth.com/insights/how-a-new-technology-add-on-payment-works>

4 5 9 10 New Medical Services and New Technologies | CMS
<https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/new-medical-services-and-new-technologies>

8 Drug and Device Manufacturer Communications With Payors, Formulary Committees, and Similar Entities – Questions and Answers | FDA

<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/drug-and-device-manufacturer-communications-payors-formulary-committees-and-similar-entities>